

Equity Volatility Strategy

Correlation and Volatility at the Lows: What to do

Despite a busy macro schedule ahead, risk premiums for the events are very subdued in [equities](#) and [beyond](#). Implied correlation and volatility are trading near the low end of their range, while spot levels are near all-time highs. This presents a good entry point for hedging in general, but we want to highlight two relative value trades in particular.

- **Carry trade that also pays in drawdowns:** Unlike most vol carry trades which suffer in large drawdowns due to harvesting the left tail, a **Knock-Out Var vs Varswap spread** is a unique payout that keeps the carry. Current market conditions are very favourable due to the high spot and low vol combination. We discuss the details of the mechanism in this note.
- **Legging in Single Stocks Vega:** Single stock implied vol reset post earnings has been significant. Implied vols are reaching a floor even on some **very cyclical names**. We like legging in some longer dated vega outright with the technique of [selling index leg later](#). Systematic dispersion investors can consider **gamma-neutral basket** accordingly.

Global Quantitative and Derivatives Strategy

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Carry Trade that Also Pays in Drawdowns

With spot hovering around the highs and implied volatility being low, hedging with put structures is currently very attractive although a **premium bleed** is always involved. Conversely, carry trades often entail significant left tail risk that can **wipe out premiums** collected during large drawdowns.

Knock-out Var vs Varswap spread is a very unique RV trade where, as long as the spot remains below the knock-out level, the carry from the vol-spread is maintained, **regardless of the magnitude of the drawdown**. The risk in this trade is to the **upside rather than the downside**, with a well-contained magnitude.

This is particularly relevant now due to two elements in the current conditions:

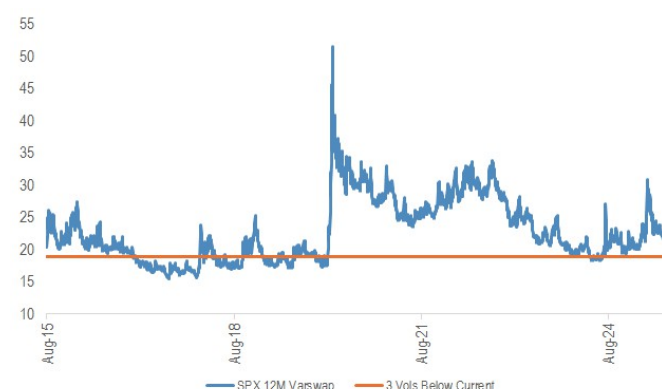
- Spot is near all-time highs with upside momentum slowing due to [overcrowding](#)
- Vanilla varswap level is near historical lows, offering limited risk of worst-case PnL

Figure 1: High Beta Crowding at ~100%ile



Source: J.P. Morgan Equity Strategy & Quant Research

Figure 2: SPX Varswap Strike Level for 12M Tenor



Source: J.P. Morgan Equity Derivatives Strategy

A Knock-out varswap is a variation of the traditional varswap that can expire early if the spot closes **above a predetermined barrier** on any daily observation date. Variance is accrued up to the day the spot knocks out (or up to maturity if no knock-out event occurs), while the vega notional is scaled by the number of days until then.

$$KO \text{ Var Payout} = 100 * \frac{VegaNotional}{2K_{ref}} * \frac{N_{KO}}{N} \left(\frac{252}{N_{KO}} \sum_{i=1}^{N_{KO}} \ln \left(\frac{S_t}{S_{t-1}} \right)^2 - K_{KO}^2 \right)$$

The knock-out feature eliminates the collection of realized volatility above the barrier, where cheap volatility exists due to skew, resulting in the **KO Var strike higher than the same tenor vanilla varswap**. While the spread is model-dependent, its value remains fairly stable for longer maturities. Consequently, **the timing of the spot entry level is very important**.

If spot remains below the barrier, **the full carry would be collected**. When a KO event occurs, the investor typically rolls the KO var into a new one. Carry accrued prior to the KO event will be collected on a “pro-rata” basis. Therefore, as long as the investor can sell the new KO var above the initial vanilla varswap strike, minus some positive buffer from carry collected, the position would not incur a loss.

This is where the second favorable condition comes into play: the **1Y varswap is at historically low levels**. Given the KO var is around 3 vol points above the vanilla varswap, rolling into a new KO var at “breakeven” would imply that the vanilla var swap would need to be around 3 vols lower from current levels, ignoring accrued carry (Figure 1). This corresponds to the low range of varswap levels seen prior to 2020. If a knock-out occurs later, there could be a risk of the roll-down, but the spread would have also accumulated a good buffer before the event.

Sell Jun26 102% SPX Index Knock-Out Var vs Var indicatively bid at 3 vols

(Vanilla Var Ref at 22)

Stocks Vols are Looking Attractive

In this earnings season, we have observed a significant vol **reset post results** across the board for single stocks, even for longer tenors. In Europe, this was associated with more spot downside movements where vol underperformed skew, while in the US, the resets were more prominent regardless. Since then, single stock volatilities have largely settled at the lower end of the range.

Figure 3: Top Earnings Vol Reset in Large Cap US and EU Names

Ticker	Move	3M	6M	1Y	Ticker	Move	3M	6M	1Y
UNH	-7.5%	-7.5%	-4.8%	-3.4%	LSEG LN	-7.9%	-3.6%	-2.4%	-1.6%
IBM	-7.6%	-5.5%	-3.7%	-2.5%	UMG NA	-5.2%	-3.8%	-2.0%	-1.3%
NFLX	-5.1%	-5.2%	-3.4%	-2.4%	RHM GY	-8.0%	-3.3%	-1.8%	-1.1%
TSLA	-8.2%	-3.5%	-2.3%	-2.3%	SAP GY	-4.1%	-3.1%	-1.7%	-1.1%
AMD	-6.4%	-5.5%	-3.4%	-2.0%	HSBA LN	-4.5%	-1.4%	-0.8%	-1.1%
NOW	4.2%	-5.1%	-2.6%	-2.0%	ABI BB	-11.6%	-1.4%	-0.4%	-0.8%
APP	12.0%	-1.8%	-4.2%	-1.9%	MUV2 GY	-7.2%	-0.7%	-0.8%	-0.8%
GE	-2.2%	-5.1%	-2.7%	-1.8%	SREN SE	-3.5%	-1.2%	-0.6%	-0.8%
PANW	3.1%	-4.3%	-2.3%	-1.7%	SU FP	-4.6%	-2.1%	-1.2%	-0.8%
MRK	-1.7%	-2.9%	-2.0%	-1.6%	DG FP	-2.6%	-1.2%	-0.5%	-0.7%

Source: J.P. Morgan Equity Derivatives Strategy

Implied correlation is also very subdued (Figure 6), indicating that index vol is not a clear sell currently for dispersion packages. Therefore, one could **begin “cherry-picking” some stock vega** to own now and consider [selling index vol later using the techniques we explored previously](#).

In the current macro backdrop, we favor owning vega in stocks that have already started showing **vulnerability in the latest earnings** within the US universe. We selected names where the implied volatility is at the low end of its historical range and filtered them by negative price reactions on earnings day.

Figure 4: Short List of US Names to Buy Vega

Jun-26 70-130 Corridor Varswap Indication

Ticker	NAME	GICS_SECTOR_NAME	Vol %tile	Eday Move	Offer
SYK	STRYKER CORP	Health Care	0.7%	-3.8%	23.45%
ABNB	AIRBNB INC-CLASS A	Consumer Discretionary	1.6%	-8.0%	35.66%
COIN	COINBASE GLOBAL INC -CLASS A	Financials	1.6%	-16.7%	57.44%
CPNG	COUPANG INC	Consumer Discretionary	1.6%	-6.1%	33.58%
PYPL	PAYPAL HOLDINGS INC	Financials	3.0%	-8.7%	35.40%
ECL	ECOLAB INC	Materials	3.1%	-3.9%	21.68%
AMT	AMERICAN TOWER CORP	Real Estate	3.7%	-4.2%	25.76%
AMZN	AMAZON.COM INC	Consumer Discretionary	4.7%	-8.3%	31.14%
PFE	PFIZER INC	Health Care	5.8%	-3.3%	25.26%
DIS	WALT DISNEY CO/THE	Communication Services	6.6%	-2.0%	28.00%
USB	US BANCORP	Financials	7.2%	-1.0%	27.51%
NFLX	NETFLIX INC	Communication Services	8.2%	-5.1%	34.34%
FCX	FREEPORT-MCMORAN INC	Materials	12.6%	-2.1%	37.40%
NEE	NEXTERA ENERGY INC	Utilities	12.6%	-6.1%	27.02%
QCOM	QUALCOMM INC	Information Technology	16.4%	-7.7%	33.11%
AMAT	APPLIED MATERIALS INC	Information Technology	21.3%	-14.1%	36.56%
BA	BOEING CO/THE	Industrials	25.7%	-4.4%	32.92%
XOM	EXXON MOBIL CORP	Energy	27.2%	-1.8%	24.65%
LRCX	LAM RESEARCH CORP	Information Technology	33.7%	-4.3%	38.91%
DE	DEERE & CO	Industrials	35.8%	-6.8%	28.07%
CARR	CARRIER GLOBAL CORP	Industrials	39.1%	-10.6%	31.16%
AMGN	AMGEN INC	Health Care	44.0%	-5.1%	27.57%
MCK	MCKESSON CORP	Health Care	45.2%	-5.8%	25.43%

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg L.P.

In Europe, with a smaller universe of liquid stocks to choose from, we have shortlisted names that exhibit low implied volatility, with a preference for stocks that are more cyclical, such as those in the **Consumer, Tech and Banks**.

Figure 5: Short List of US Names to Buy Vega

Jun-26 70-130 Corridor Varswap Indication

Ticker	NAME	GICS_SECTOR_NAME	Vol %tile	Offer
TTE FP	TOTALENERGIES SE	Energy	1.1%	21.1%
LONN SE	LONZA GROUP AG-REG	Health Care	2.3%	26.2%
PHIA NA	KONINKLIJKE PHILIPS NV	Health Care	10.1%	31.9%
UBSG SE	UBS GROUP AG-REG	Financials	13.4%	27.7%
RNO FP	RENAULT SA	Consumer Discretionary	18.8%	33.5%
INGA NA	ING GROEP NV	Financials	20.0%	26.5%
LLOY LN	LLOYDS BANKING GROUP PLC	Financials	24.3%	26.9%
HSBA LN	HSBC HOLDINGS PLC	Financials	25.1%	23.3%
BARC LN	BARCLAYS PLC	Financials	25.4%	30.8%
ASML NA	ASML HOLDING NV	Information Technology	28.8%	34.5%
ACA FP	CREDIT AGRICOLE SA	Financials	29.9%	24.1%
VOW3 GY	VOLKSWAGEN AG-PREF	Consumer Discretionary	31.0%	29.1%
ISP IM	INTESA SANPAOLO	Financials	31.3%	26.1%
RMS FP	HERMES INTERNATIONAL	Consumer Discretionary	32.3%	27.3%
CFR SE	CIE FINANCIERE RICHEMO-A REG	Consumer Discretionary	37.1%	29.9%

\$50k indicative size

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg L.P.

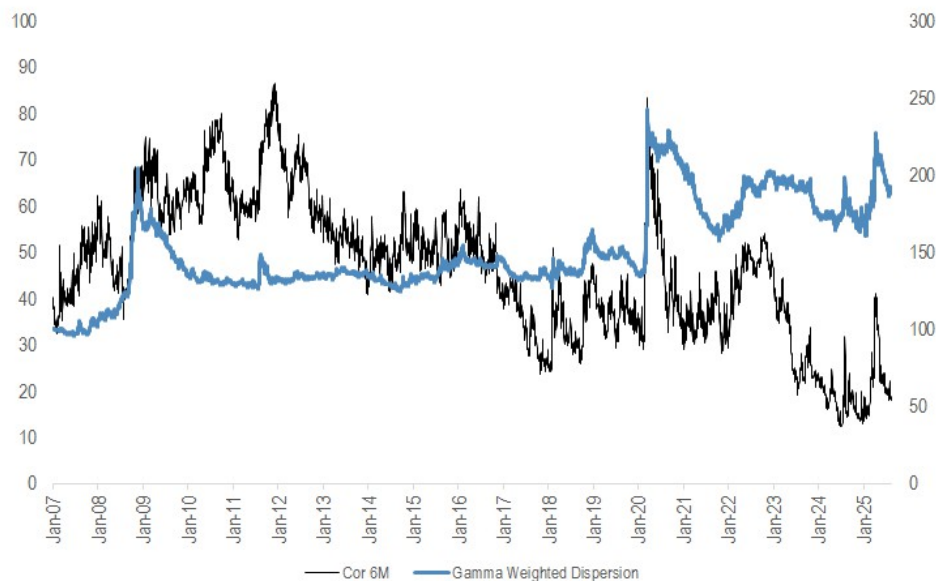
For investors looking to trade dispersion systematically, a **gamma-weighted dispersion strategy** is worth considering. We previously discussed the [issue of being net short gamma](#) even in a vega-neutral dispersion package, where **low implied correlation** can amplify the MTM pain during a sharp correlation-one drawdown event. A gamma-neutral dispersion would mitigate this problem by having more vega on the single stock leg. We have plotted a 6M model strategy performance against implied correlation,

demonstrating that entering the trade when correlation is at its lows has historically been beneficial.

Figure 6: Gamma-Weighted Dispersion on SPX T75 6M Tenor

Left axis - 6M Implied Correlation

Right axis - Gamma Weighted Dispersion Portfolio 6M 95-105 Strangles



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg L.P.

Appendix - Related Publications

Recent Publications from the Equity Volatility Strategy

[Thinking Outside the Box for Tail Trading](#)

[Thinking Outside the Box for Tail Trading - Part 2](#)

[Dispersion Update, Index Technique, Stock Leg and What Now](#)

[Trading US macro events with short-dated options](#)

[The case for long dated Upvar](#)

Past Key Reports on Dispersion

Comprehensive Updates

Sep 2023: [A Practical Dispersion Trading Handbook](#)

Sep 2024: [Systematic Dispersion Update and Trade Ideas](#)

Apr 2025: [Dispersion Update, Index Technique, Stock Leg and What Now](#)

Exploration on Topics

Aug 2022: [Deep dive into single stock volatility behavior around earnings \(EU\)](#)

Oct 2022: [Deep dive on stock volatility behavior around earnings \(US\)](#)

May 2023: [Patience is a virtue: delaying index vol selling in dispersion trades](#)

Jul 2023: [Assessing extreme low correlation, Index and Stock Vol Ratio Study](#)

Mar 2024: [Taking Stock of Record Low Correlation](#)

Apr 2024: [Deep Dive on Index Call Strategies](#)

May 2024: [Autos Sector Volatility Trade \(corridor varswap vs volswap\)](#)

May 2024: [Play further tech earnings with index and earnings vol reset recap](#)

Feb 2025: [Buy correlation with VSTOXX](#)

Mar 2025: [The Call for Call Overwriting \(Use stock upside to buy leveraged index leg\)](#)

Mar 2025: [Taking Stock of the S&P 500 Correlation Pop](#)

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